On-board SLAM payload dashboard

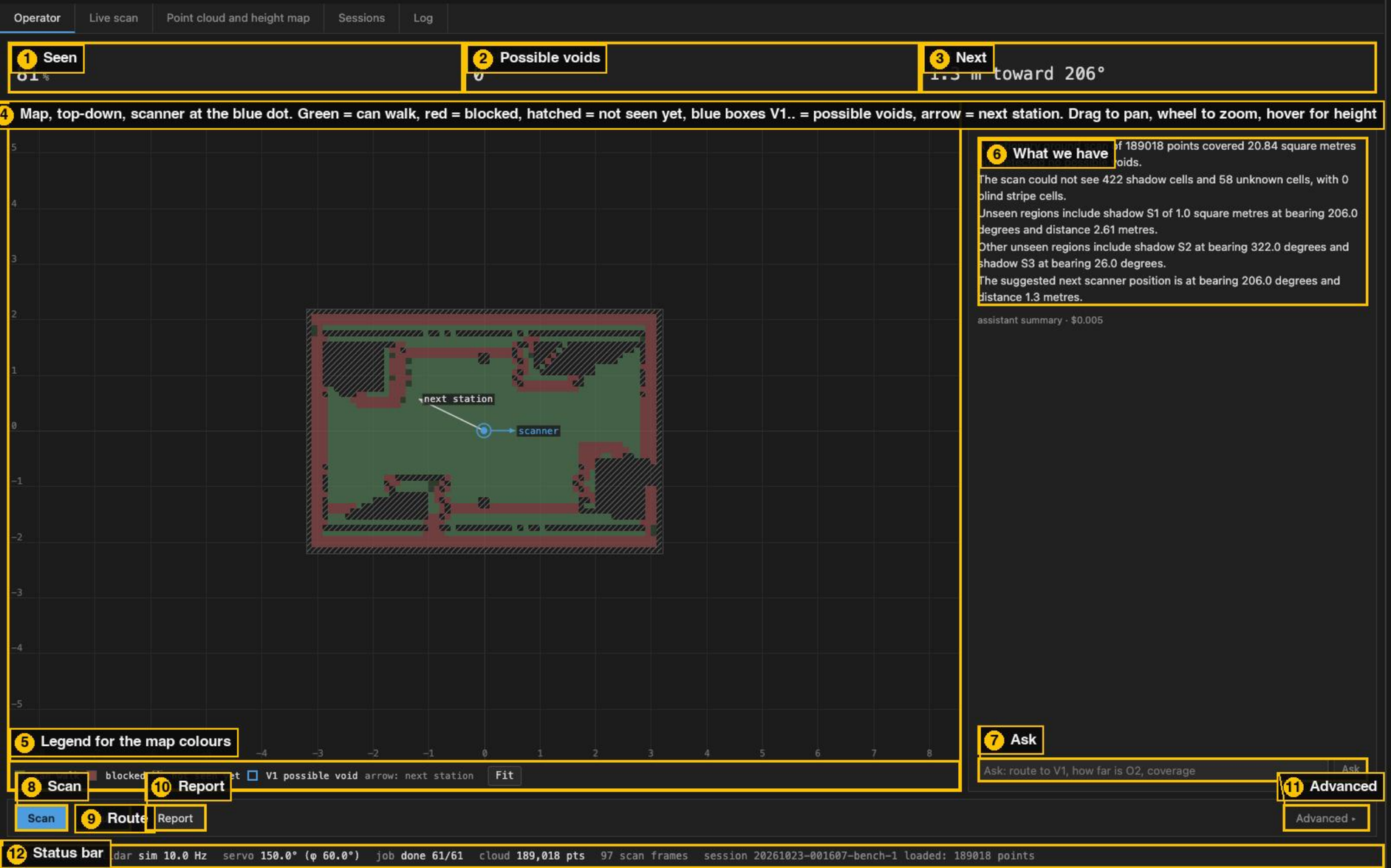
What each part of each screen does

Screenshots from the running tool on the simulator, 22 Oct 2026.
Every screen has the same shape: numbers on top, one main view, one column in words, buttons at the bottom, and an Advanced drawer for engineering detail.

Pages: Operator (and its Advanced drawer), Live scan (and Advanced), Point cloud and height map, Sessions, Log.

Yellow outlines mark the parts; the numbered label on each part matches the explanation in the column on the right of the page.

Operator: the screen for the person at the deployment point



1. Seen: how much of the area the scanner could actually see, in %
2. Possible voids: cells where something sits over open space a person could fit in. Says nothing about whether anyone is there
3. Next: where to carry the scanner to uncover the largest hidden area (distance and compass-style bearing from the scanner)
4. Map, top-down, scanner at the blue dot. Green = can walk, red = blocked, hatched = not seen yet, blue boxes V1.. = possible voids, arrow = next station. Drag to pan, wheel to zoom, hover for height
5. Legend for the map colours
6. What we have: the situation written from the measured numbers after every scan. 'assistant summary' = Gemini wrote it; otherwise a fixed template. It never claims a person is present
7. Ask: plain questions such as 'can we get to V1?' or 'what did we miss?'. Jev reads the question, code computes the answer, Gemini phrases it; a route answer is drawn on the map
8. Scan: run a tilt sweep with the current settings; progress shows here
9. Route: click a start, then a goal; shows length, passable or not, worst step
10. Report: downloads the map as PNG and the findings as JSON
11. Advanced: layers, limits, full lists, ranking, scan settings
12. Status bar: connection, LiDAR rate, servo angle, job state, cloud size, last message, alarms in red

Operator: the Advanced drawer

Operator | Live scan | Point cloud and height map | Sessions | Log

Seen 81 %

Possible voids 0

Next 1.3 m toward 206°



What we have 00:18:41

The scan covered 20.84 square metres and detected no possible voids.
The scanner could not see 422 shadow cells and 58 unknown cells.
Unseen regions include shadow S1 of 1.0 square metre at bearing 206.0 degrees and distance 2.61 metres.
Other unseen regions include shadow S2 at bearing 322.0 degrees and distance 2.19 metres, and shadow S3 at bearing 26.0 degrees and distance 2.71 metres.
The suggested next scanner position is at bearing 206.0 degrees and distance 1.3 metres.

assistant summary · $0.005

Ask: route to V1, how far is O2, coverage — Ask

Scan | Route | Report — Advanced ▾

1 Advanced drawer shadow blind findings next station | Cell m 0.1 | Max step m 0.15 | Refresh analysis | Fit — 64×44 @ 0.1 m · 00:18:41 · job done 61/61

Scan job done

Job	#1 bench-1
Step	61 of 61
φ now	60.0°
Points	189,018
ETA	–

Voids 0

none

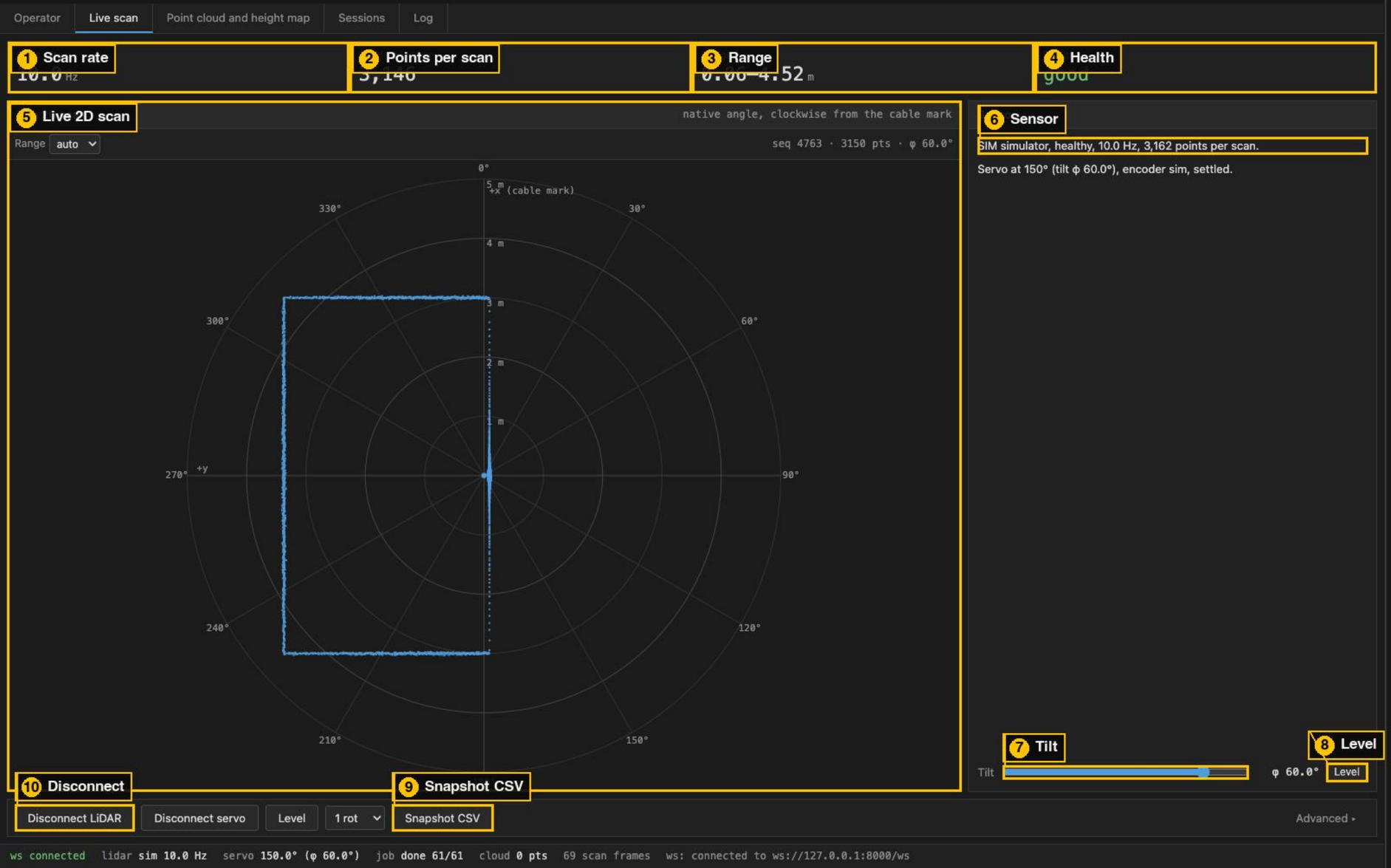
Obstacles 7

id	m²	height m	dist m	bearing
01	3.12	2.99	1.68	270°

ws connected lidar sim 10.0 Hz servo 150.0° (φ 60.0°) job done 61/61 cloud 189,018 pts 96 scan frames session 20261023-001607-bench-1 loaded: 189018 points

1 Advanced drawer: map layer toggles (terrain, shadow vs blind vs unknown), cell size and step/slope limits, the full lists of obstacles, voids, depressions and shadows (click one to highlight it), Jev's ranking with confidence bars and the rescan verdict, and the scan job settings (tilt range, step, rotations, settle time)

Live scan: is the hardware alive, which way is it pointing



1. Scan rate: rotations per second (the S2 does 10)
2. Points per scan: returns in the last rotation (about 3,200 on the S2)
3. Range: nearest and farthest return in the last rotation
4. Health: reported by the LiDAR itself; 'off' when not connected
5. Live 2D scan: one rotation of the LiDAR as a polar plot. 0° at the cable mark, clockwise. This is what the scan plane hits at the current tilt; tilt it and the shape changes live. Range rings in metres
6. Sensor: LiDAR and servo state in words (model, port, health, rate; servo angle, tilt ϕ, encoder, settled). When something is disconnected the port picker and Connect button appear here
7. Tilt: drag to tilt the cradle; ϕ is the scan-plane angle, 0 = horizontal
8. Level: send the servo to its centre so the scan plane is horizontal
9. Snapshot CSV: save the next N raw rotations (angle, range, quality) for calibration and bias measurements
10. Disconnect: release the port (LiDAR / servo)

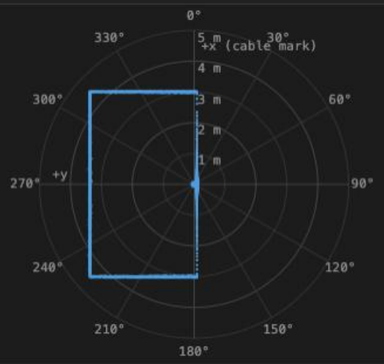
Live scan: the Advanced drawer

Operator | **Live scan** | Point cloud and height map | Sessions | Log

Scan rate	Points per scan	Range	Health
10.0 Hz	3,159	0.06–4.53 m	good

Live 2D scan — native angle, clockwise from the cable mark

Range: auto — seq 4836 · 3159 pts · φ 60.0°



Sensor

SIM simulator, healthy, 10.0 Hz, 3,159 points per scan.

Servo at 150° (tilt ϕ 60.0°), encoder sim, settled.

Tilt — φ 60.0° — Level

Disconnect LiDAR | Disconnect servo | Level | 1 rot | Snapshot CSV | Advanced

1 Advanced drawer — 3159 of 3159 rows

Angle filter: from — to

#	angle °	range m	quality
0	0.00	3.008	40
1	0.11	3.018	46
2	0.22	2.979	58
3	0.34	3.030	57
4	0.45	2.995	57
5	0.56	3.006	40
6	0.68	2.995	44
7	0.79	2.988	57
8	0.90	3.016	59
9	1.01	2.997	52
10	1.13	2.986	48
11	1.24	2.999	40
12	1.35	2.996	57

Servo console — 150.0° servo · φ 60.0°

Tilt ϕ (deg): 60.0 — Go

Mapping: ϕ = (servo − 90)

Raw console (calibration): ? — Send

Commands: ? status, I identify, T deg, S deg ms, H home, L min max, C us_min us_max range

Connection

Lidar

State	connected (sim)
Model	SIM
Serial	SIMULATED
Firmware	0.0 / hw 0
Health	good
Scan rate	10.0 Hz

Mount and filters

Servo to tilt

Centre (servo °): 90 — Use current

Invert — ϕ = (servo − 90)

Encoder to tilt (AS5600)

Centre (meas °): 90 — Use measured

Invert — encoder sim

Geometry (mm, deg)

Axis height: 80

h: plane − axis: 0

Lidar yaw: 0

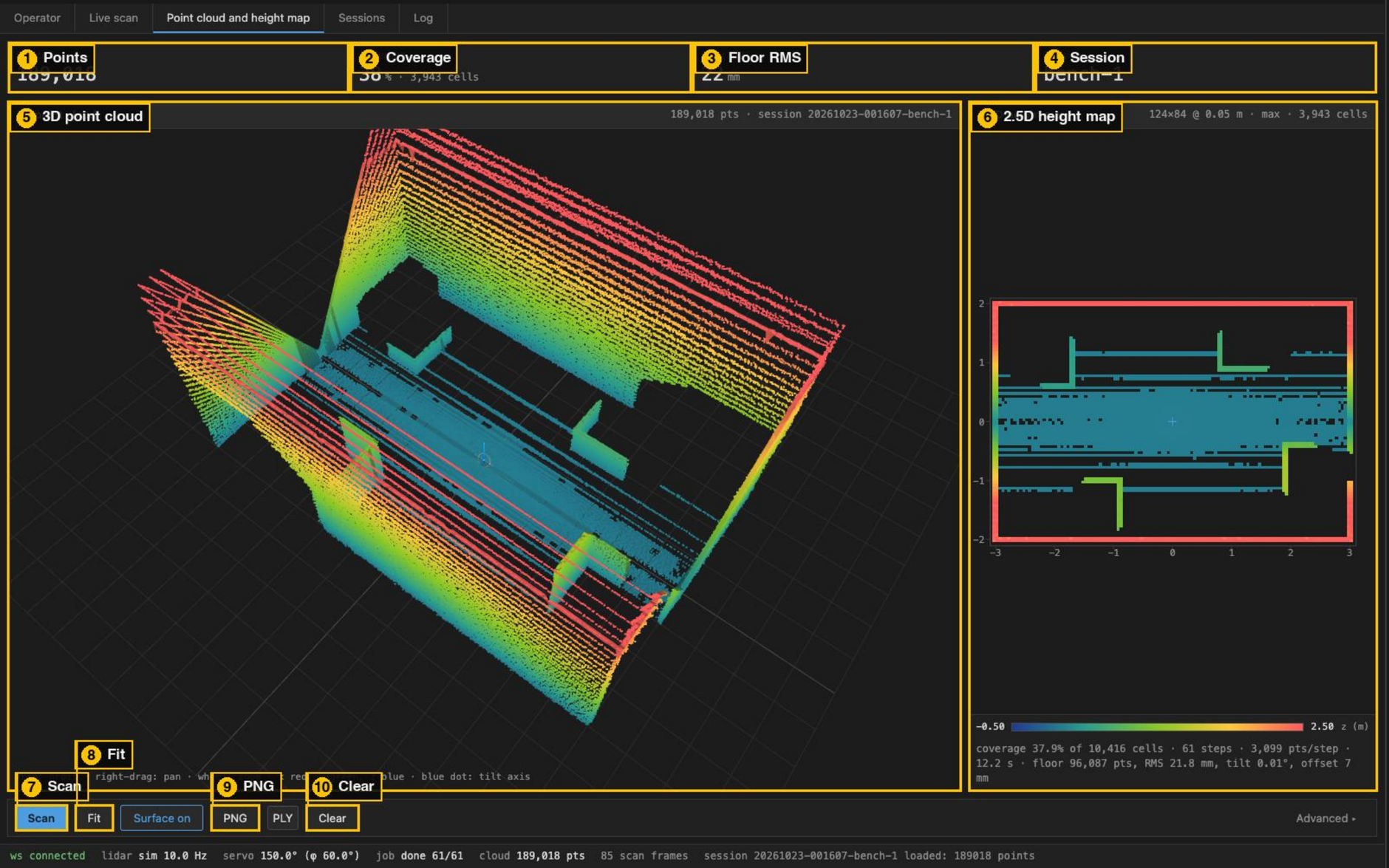
Point filters (next sweep)

Min quality: 10

ws connected | lidar sim 10.0 Hz | servo 150.0° (φ 60.0°) | job done 61/61 | cloud 0 pts | 69 scan frames | ws: connected to ws://127.0.0.1:8000/ws

1 Advanced drawer: the raw values table (every return of the last rotation, angle filter), the servo console for calibration commands (? status, T angle, W wait for settle, E encoder diagnostics, C pulse calibration), connection details (model, serial, firmware, port, baud), alarms, and Mount and filters (servo and encoder centre, invert, axis height, scan-plane offset h, lidar yaw, point quality and range limits)

Point cloud and height map: the raw 3D scan and its 2.5D reduction



1. Points: in the current cloud (live sweep or loaded session)
2. Coverage: filled cells of the height grid's bounding box
3. Floor RMS: plane fit to points within 10 cm of the floor; the scanner's noise figure
4. Session: which saved sweep is shown, or 'live'
5. 3D point cloud: every tilt step draws one slice of the room; colour = height (blue floor, red high). Drag to orbit, right-drag to pan, wheel to zoom. The grey sheet is the 2.5D surface
6. 2.5D height map: the same scene reduced to one height per 5 cm cell, seen from above. Under it: coverage, steps, points per step, sweep time, floor fit
7. Scan: run a sweep
8. Fit: reset the camera to the overhead view
9. PNG: save the height map image for the report
10. Clear: drop the current cloud

Sessions: every saved sweep

Operator | Live scan | Point cloud and height map | Sessions | Log

Sessions | Total points | Loaded

1 Saved sweeps (raw angles, ranges, tilt per point, so they can be re-processed later). Load one into the views, export PLY or CSV, delete

Sessions loaded: 20261023-001607-bench-1

Id	Name	Created	Points	Steps	φ range	Lidar	Actions
20261023-001607-bench-1	bench-1	23/10/2026, 00:16:07	189,018	61	-60° to 60° by 2° × 1	SIM	Load Grid PLY CSV Delete
20261022-234351-soak-std	soak-std	22/10/2026, 23:43:51	189,036	61	-60° to 60° by 2° × 1	SIM	Load Grid PLY CSV Delete
20261022-234338-soak-fine	soak-fine	22/10/2026, 23:43:38	375,586	121	-60° to 60° by 1° × 1	SIM	Load Grid PLY CSV Delete
20261022-234314-soak-wide	soak-wide	22/10/2026, 23:43:14	229,073	76	-75° to 75° by 2° × 1	SIM	Load Grid PLY CSV Delete
20261022-234258-soak-double	soak-double	22/10/2026, 23:42:58	195,106	31	-45° to 45° by 3° × 2	SIM	Load Grid PLY CSV Delete
20261022-234248-soak-std	soak-std	22/10/2026, 23:42:48	189,111	61	-60° to 60° by 2° × 1	SIM	Load Grid PLY CSV Delete
20261022-224832-ops-check	ops-check	22/10/2026, 22:48:32	189,089	61	-60° to 60° by 2° × 1	SIM	Load Grid PLY CSV Delete
20261022-171742-shot	shot	22/10/2026, 17:17:42	40,940	13	-30° to 30° by 5° × 1	SIM	Load Grid PLY CSV Delete
20261022-171651-shot	shot	22/10/2026, 17:16:51	40,956	13	-30° to 30° by 5° × 1	SIM	Load Grid PLY CSV Delete
20261022-165527-smoke	smoke	22/10/2026, 16:55:27	59,802	19	-45° to 45° by 5° × 1	SIM	Load Grid PLY CSV Delete

Saved sweeps

10 saved sweeps with 1,697,717 points in total.

Loaded: bench-1 (189,018 points, 61 steps).

Load puts a session into the live view (point cloud, height map, Operator). Grid only fetches its height map. PLY and CSV download the points.

Refresh

ws connected lidar sim 10.0 Hz servo 150.0° (φ 60.0°) job done 61/61 cloud 0 pts 59 scan frames ws: connected to ws://127.0.0.1:8000/ws

1 Saved sweeps (raw angles, ranges, tilt per point, so they can be re-processed later). Load one into the views, export PLY or CSV, delete

Log: what the server and the dashboard said

Operator | Live scan | Point cloud and height map | Sessions | Log

Errors	Warnings	Entries
0	0	1

1 **Messages from the server and the dashboard in time order; warnings and errors in colour; alarms raised and cleared appear here**

```
00:19:18  info  client  ws: connected to ws://127.0.0.1:8000/ws
```

Latest

No errors.

No warnings.

Server messages come over the WebSocket; client lines are this dashboard's own calls.

Clear

ws connected lidar sim 10.0 Hz servo 150.0° (φ 60.0°) job done 61/61 cloud 0 pts 59 scan frames ws: connected to ws://127.0.0.1:8000/ws

1 Messages from the server and the dashboard in time order; warnings and errors in colour; alarms raised and cleared appear here